COURSE CODE & NAME : CSA1712 ARTIFICAL INTELLIGENCE.

EXP 16 : Feed Forward Neural Network (Python)

main.py | Visualize | Run

```python
import numpy as np
def sigmoid(x):
    return 1 / (1 + np.exp(-x))
def sigmoid_derivative(x):
    return x * (1 - x)
X = np.array([[0, 0],
              [0, 1],
              [1, 0],
              [1, 1]])
y = np.array([[0], [1], [1], [0]])
np.random.seed(42)
input_neurons = 2
hidden_neurons = 2
output_neurons = 1
weights_input_hidden = np.random.uniform(size=
(input_neurons, hidden_neurons))
bias_hidden = np.random.uniform(size=(1,
hidden_neurons))
weights_hidden_output = np.random.uniform(size=
(hidden_neurons, output_neurons))
bias_output = np.random.uniform(size=(1,
output_neurons))
epochs = 10000
```

Output

```
Output after Feed Forward Training:
[[0.0532]
 [0.9517]
 [0.9516]
 [0.0518]]
```

Microsoft Edge

EXP 17 : Sum of Integers from 1 to N (Prolog)



EXP 18 : Database with Name and DOB (Prolog)

```
1
2 dob(raj, date(2004, 5, 12)).
3 dob(anu, date(2005, 8, 15)).
4 dob(ravi, date(2003, 1, 20)).
5 dob(priya, date(2004, 11, 10)).
6
```

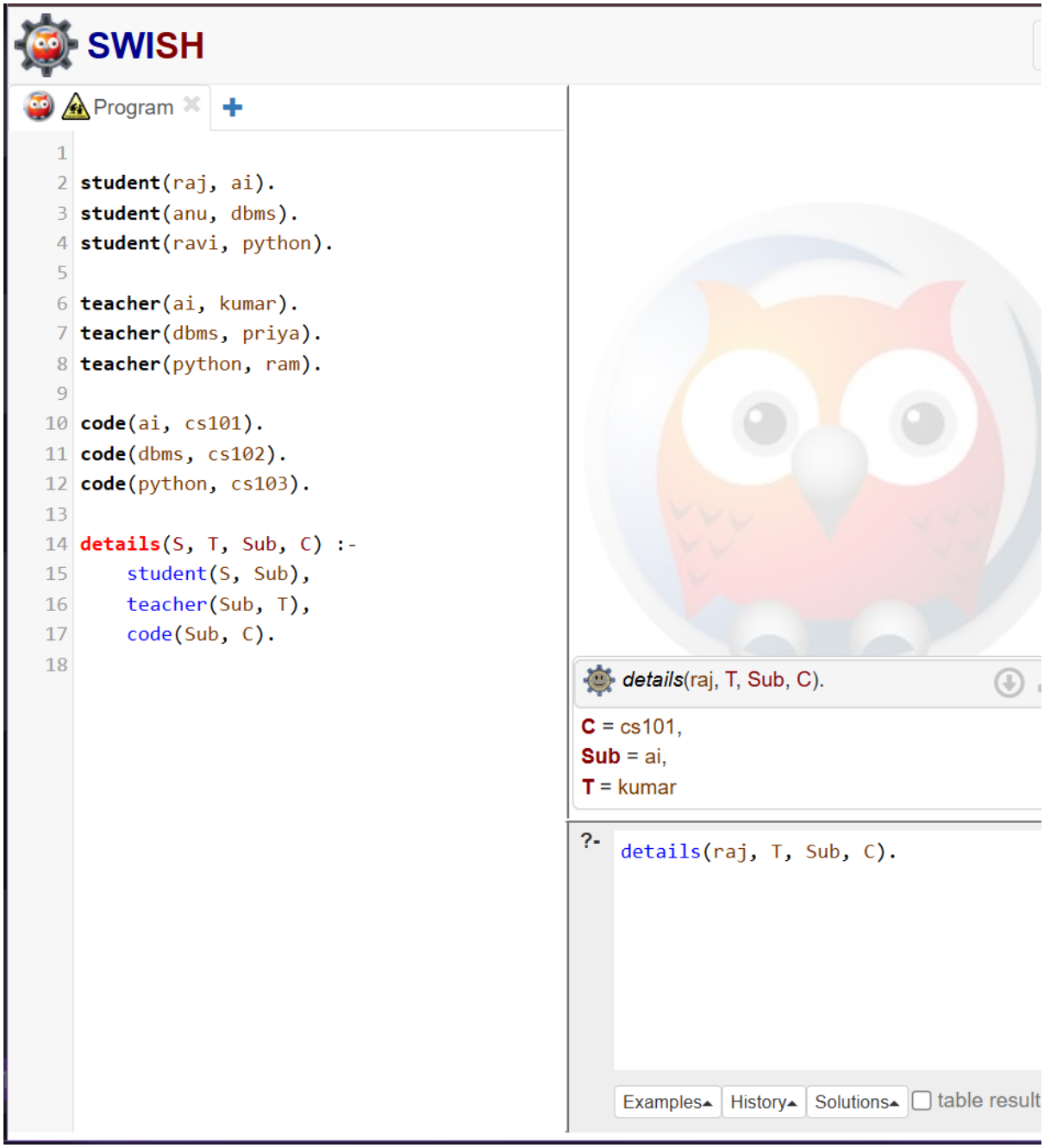


```
dob(anu, X).
X = date(2005,8,15)

?- dob(anu, X).
```

Examples▲ History▲ Solutions▲ table results

EXP 19 : Student–Teacher–Subject–Code (Prolog)



EXP 20 : Planets Database (Prolog)

